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Cost of the system

Installation costs

Item	Quantity units	Cost Ksh	Total Ksh
PV modules			
JKM-585N-72HL4-BDV	154	11,000.00	1,694,000.00
Supports for modules	154	500.00	77,000.00
Batteries	8	85,000.00	680,000.00
Controllers			35,000.00
Other components			
Accessories, fasteners	2500	1.00	2,500.00
Wiring	40000	1.00	40,000.00
Combiner box	20000	1.00	20,000.00
Monitoring system, display screen	5000	1.00	5,000.00
Measurement system, pyranometer	5000	1.00	5,000.00
Surge arrester	2500	1.00	2,500.00
Installation			
Global installation cost per module	154	500.00	77,000.00
Global installation cost per inverter	1	1,000.00	1,000.00
Global installation cost per battery	8	1,000.00	8,000.00
Settings	1	5,000.00	5,000.00
Taxes			
VAT	1	0.00	28,160.00
		Total	2,680,160.00
		Depreciable asset	2,488,500.00

Operating costs

Item	Total Ksh/year
Maintenance	
Repairs	150,000.00
Cleaning	20,000.00
Provision for battery replacement	200,000.00
Security fund	50,000.00
Total (OPEX)	420,000.00
Including inflation (8.00%)	961,001.25

System summary

Total installation cost	2,680,160.00	Ksh
Operating costs (incl. inflation 8.00%/year)	961,001.25	Ksh/year
Excess energy (battery full)	56.1	MWh/year
Used solar energy	104	MWh/year
Used energy cost	10.5034	Ksh/kWh

Financial analysis

Simulation period

Project lifetime	20 years	Start year	2027

Income variation over time

Inflation	8.00 %/year
Production variation (aging)	0.00 %/year
Discount rate	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (Ksh)	Depreciable (Ksh)
PV modules				
JKM-585N-72HL4-BDV	Straight-line	20	0.00	1,694,000.00
Supports for modules	Straight-line	20	0.00	77,000.00
Batteries	Straight-line	20	0.00	680,000.00
Controllers	Straight-line	20	0.00	35,000.00
Accessories, fasteners	Straight-line	20	0.00	2,500.00
		Total	0.00	2,488,500.00

Financing

Own funds	2,680,160.00 Ksh

Electricity sale

Feed-in tariff	25.00000 Ksh/kWh

Return on investment

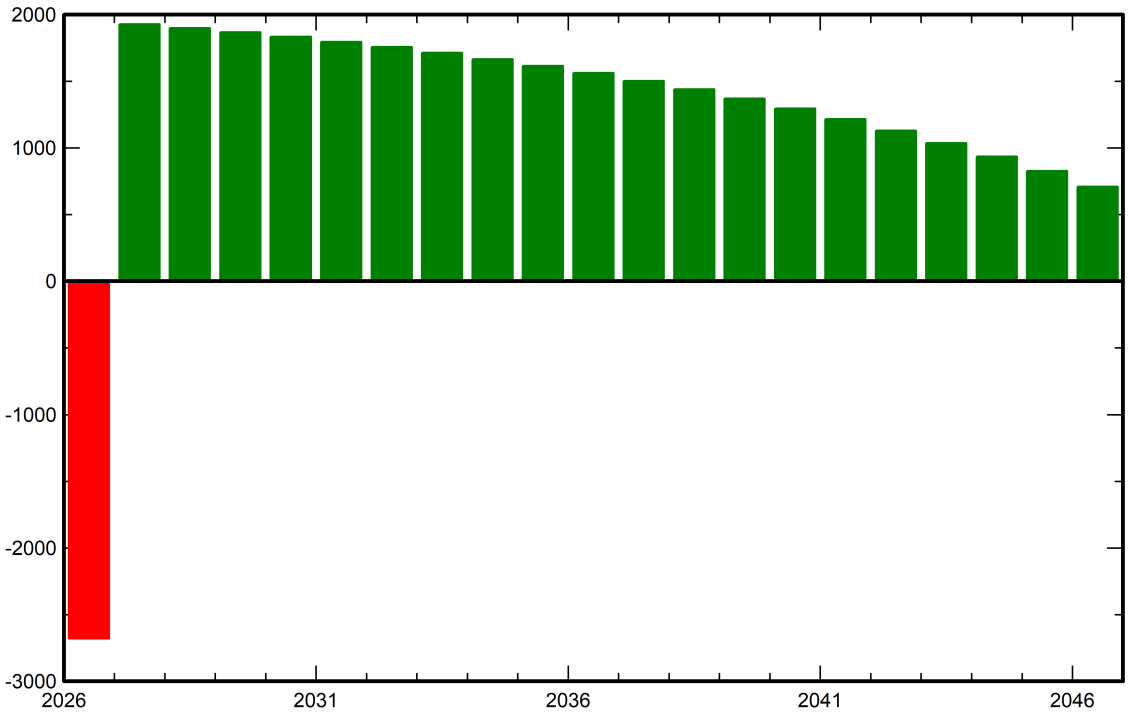
Payback period	1.4 years
Net present value (NPV)	26,423,874.67 Ksh
Internal rate of return (IRR)	70.20 %
Return on investment (ROI)	985.9 %

Financial analysis

Detailed economic results (kKsh)

Year	Electricity sale	Own funds	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	2,680,160	0	0	0	0	0	-2,680,160	0.0%
1	2,606,314	0	420,000	124,425	2,061,889	257,736	1,928,578	-751,582	72.0%
2	2,606,314	0	453,600	124,425	2,028,289	253,536	1,899,178	1,147,596	142.8%
3	2,606,314	0	489,888	124,425	1,992,001	249,000	1,867,426	3,015,021	212.5%
4	2,606,314	0	529,079	124,425	1,952,810	244,101	1,833,134	4,848,155	280.9%
5	2,606,314	0	571,405	124,425	1,910,484	238,810	1,796,098	6,644,253	347.9%
6	2,606,314	0	617,118	124,425	1,864,771	233,096	1,756,100	8,400,353	413.4%
7	2,606,314	0	666,487	124,425	1,815,402	226,925	1,712,902	10,113,255	477.3%
8	2,606,314	0	719,806	124,425	1,762,083	220,260	1,666,247	11,779,502	539.5%
9	2,606,314	0	777,391	124,425	1,704,498	213,062	1,615,861	13,395,363	599.8%
10	2,606,314	0	839,582	124,425	1,642,307	205,288	1,561,444	14,956,807	658.1%
11	2,606,314	0	906,748	124,425	1,575,140	196,893	1,502,673	16,459,479	714.1%
12	2,606,314	0	979,288	124,425	1,502,601	187,825	1,439,200	17,898,680	767.8%
13	2,606,314	0	1,057,631	124,425	1,424,257	178,032	1,370,650	19,269,330	819.0%
14	2,606,314	0	1,142,242	124,425	1,339,647	167,456	1,296,616	20,565,946	867.3%
15	2,606,314	0	1,233,621	124,425	1,248,268	156,033	1,216,659	21,782,606	912.7%
16	2,606,314	0	1,332,311	124,425	1,149,578	143,697	1,130,306	22,912,911	954.9%
17	2,606,314	0	1,438,896	124,425	1,042,993	130,374	1,037,044	23,949,955	993.6%
18	2,606,314	0	1,554,008	124,425	927,881	115,985	936,321	24,886,276	1028.5%
19	2,606,314	0	1,678,328	124,425	803,561	100,445	827,541	25,713,817	1059.4%
20	2,606,314	0	1,812,594	124,425	669,295	83,662	710,058	26,423,875	1085.9%
Total	**52,126,279**	**2,680,160**	**19,220,025**	**2,488,500**	**30,417,754**	**3,802,219**	**29,104,035**	**26,423,875**	**1085.9%**

Yearly net profit (kKsh)



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Financial analysis

Cumulative cashflow (kKsh)

